or

$$\langle \mathbf{L} \rangle = \langle A|\mathbf{L}|A\rangle.$$

In other words, **L** has the same expectation value in state |*B*> as it does in state |*A*>. Promise kept.

4.9 CONNECTIONS TO CLASSICAL MECHANICS

The average, or expectation value, of an observable is the closest thing in quantum mechanics to a classical value. If the probability distribution for an observable is a nice bell-shaped curve, and not too broad, then the expectation value really is the value that you expect to measure. If a system is so big and heavy that quantum mechanics is not too important, then the expectation value of an observable behaves almost exactly according to classical equations of motion. For this reason, it is interesting and important to find out how expectation values change with time.

First of all, *why* do they change with time? They change with time because the state of the system changes with time. Suppose the state at time *t* is represented by ket |Ψ(*t*)> and bra <Ψ(*t*)|. The expectation value of the observable **L** at time *t* is

$$\langle \Psi(t)|\mathbf{L}|\Psi(\mathrm{t})\rangle.$$

Let's see how this changes by differentiating it with respect to *t* and using the Schrödinger equation for the time derivatives of |Ψ(*t*)> and <Ψ(*t*)|. Using the product rule for derivatives, we find that

$$\frac{d}{dt}\langle \Psi(t)|\mathbf{L}|\Psi(\mathrm{t})\rangle = \langle \dot{\Psi}(\mathrm{t})|\mathbf{L}|\Psi(\mathrm{t})\rangle + \langle \Psi(\mathrm{t})|\mathbf{L}|\dot{\Psi}(t)\rangle,$$

where, as usual, the dot means time derivative. **L** itself has no explicit time dependency, so it just comes along for the ride. Now, plugging in the bra and